Date
15/09/2023 Mathematics

Section A

1. 3) 3
2. 3) tan 60°
3. 2) 30°
4. 1) 10cm
5. 4) a=0
6. 4) Four decimal places
7. 3) 3 and 1
8. 2) (0, -1)
9. 4) No solution
10. 1) k= -9/2
11. 1) $k \neq -3$
12. 1) 8
13. 1) similar
14. 1) $x^3 y^4 z^4$
15. 3) $x^2/2 - x/2 - 6$
16. 3) intersecting or coincident
17. 1) Parallel
18. 4) 16 : 81
19. 3) Assertion (A) is true but Reason (R) is false
20. 1) Both A and R are true and R is correct explanation of A

Section B

21. $6\times5\times4\times3\times2\times1+5 \Rightarrow 6\times4\times3\times2\times1\times5(1+1)$
$= 6\times4\times3\times2\times1\times(2)\times5$
Since it has more than two factors it is composite.

22. $4x + 6y = 7 \Rightarrow 4x + 6y - 7 = 0 \quad (a_1x + b_1y + c_1 = 0)$
$kx + 18y = 21 \Rightarrow kx + 18y - 21 = 0 \quad (a_2x + b_2y + c_2 = 0)$
for infinite solutions $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$
So,

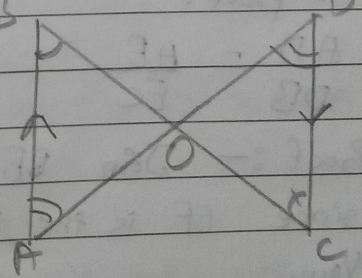
$$\frac{4}{k} = \frac{6}{18} = \frac{7}{21} \Rightarrow \frac{4}{k} = \frac{1}{3} \Rightarrow k = 12$$

$\therefore$ Value of k is 12.

23. origin (0,0) and point 2 is (-6,8)
By Distance formula $= \sqrt{(x_1-x_2)^2 + (y_1-y_2)^2}$
$\Rightarrow \sqrt{(0+6)^2 + (0-8)^2}$
$= \sqrt{36 + 64}$
$= \sqrt{100} = 10$
So, distance is 10 units.

24. Given: $\sin 3A = \cos(A - 10°)$
we know that $\sin\theta = \cos(90 - \theta)$
So, $\sin 3A = \cos(90 - 3A)$
$90 - 3A = A - 10$ (from given)
$100 = 4A \Rightarrow A = 25°$ and $3A = 75°$
So, 3A is an acute angle.

25. Given: $AB \parallel CD$
$\angle B = \angle C$ (Alternate Angle)
$\angle A = \angle D$ (Alternate Angle)
Proved above information for
ΔOAB and Δ ODC
By AA rule
$\therefore \Delta AOB \sim \Delta DOC$



$\left\{ \begin{array}{l} \Delta OAB \text{ is same} \\ \text{as } \Delta AOB \text{ and } \Delta ODC \\ \text{is same as } \Delta DOC \end{array} \right\}$

Section C

26. $\frac{\cos(90-\theta)}{1+\sin(90-\theta)} + \frac{1+\sin(90-\theta)}{\cos(90-\theta)} = 2\operatorname{cosec}\theta$

$= \frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta}$ $\left\{ \begin{array}{l} \sin(90-\theta) = \cos\theta \\ \cos(90-\theta) = \sin\theta \end{array} \right\}$

$= \frac{\sin^2\theta + (1+\cos\theta)^2}{\sin\theta(1+\cos\theta)}$

$= \frac{1-\cos^2\theta + 1+\cos^2\theta + 2\cos\theta}{\sin\theta(1+\cos\theta)}$ $\left\{ \begin{array}{l} \sin^2\theta = 1-\cos^2\theta \\ (\sin^2\theta + \cos^2\theta = 1 \end{array} \right\}$

$= \frac{2+2\cos\theta}{\sin\theta(1+\cos\theta)} \Rightarrow \frac{2(1+\cos\theta)}{\sin\theta(1+\cos\theta)} = \frac{2}{\sin\theta}$

$= 2\operatorname{cosec}\theta$ $\{ 1/\sin\theta = \operatorname{cosec}\theta)$

Hence Proved

27. Base Proportionality Theorem states that, if a line is parallel to a side of a triangle which intersects the other sides into two distinct points, then the line divides those sides in proportion for that triangle.

Let ΔABC, where line l is parallel to BC, intersect AB at D and AC at E

To prove:

$\frac{AD}{DB} = \frac{AE}{EC}$

Proof :- Join BE, CD and Draw $EF \perp AB$, $DG \perp CA$

since EF is the height of ΔADE and ΔDBE

Now,

$Ar(\Delta ADE) = 1/2(\text{Base} \times \text{Height}) = 1/2 \times AD \times EF$

$Ar(\Delta DBE) = 1/2(\text{Base} \times \text{Height}) = 1/2 \times DB \times EF$

$$\frac{ar(\triangle ADE)}{ar(\triangle DBE)} = \frac{1/2 \times AD \times EF}{1/2 \times DB \times EF} = \frac{AD}{DB} \quad \text{--- (i)}$$

Also,

$$\frac{ar(\triangle ADE)}{ar(\triangle DCE)} = \frac{1/2 \times AE \times DG}{1/2 \times EC \times DG} = \frac{AE}{EC} \quad \text{--- (ii)}$$

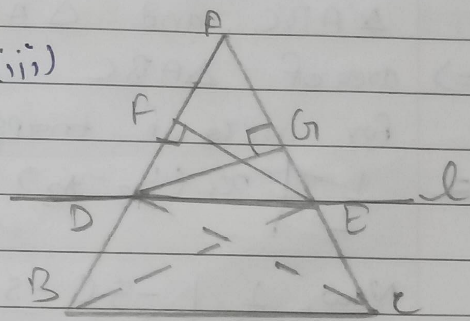
But $\triangle DBE$ and $\triangle DCE$ are on the same base DE and between the same parallel

$\therefore$ $ar(\triangle DBE) = ar(\triangle DCE)$ ---(iii)

From (i), (ii) and (iii) we have

$$\frac{AD}{DB} = \frac{AE}{EC}$$

Hence Proved



28. let Sagar age be x and Tiru age be y

Given

$x - 5 = 2(y-5)$ --- (i)

$x + 10 - (y+10) = 10$ --- ii

So, $x - 5 = 2y - 10 \Rightarrow x - 2y = -5$

$x + 10 - y - 10 = 10 \Rightarrow x - y = 10$

Now, By elimination

$$\begin{array}{r} x - 2y = -5 \\ -\quad x - y = 10 \\ \hline -y = -15 \end{array} \Rightarrow y = 15$$

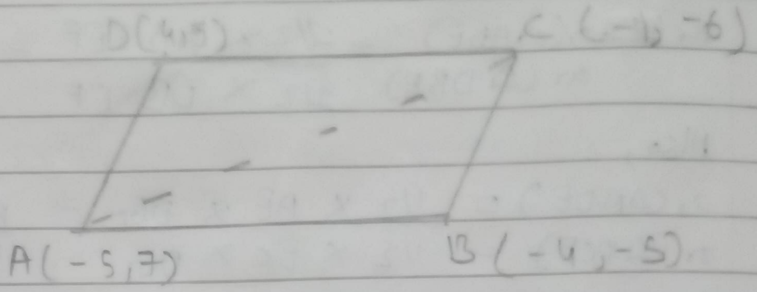
Now, $x - 2(15) = -5 = x - 30 = -5 \Rightarrow x = 25$

So, age of Sagar is 25 years

age of Tiru is 15 years

When Tiru have borned Sagar's age was 10 years.

29. Given



A (-5,7)
B (-4,-5)
C (-1,-6)
D (4,5)

To find its area we will spilt it into 2 triangles
ΔABC and ΔACD

=> area of ΔABC
for area of triangle we have :-

$$= \frac{1}{2} \left| x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2) \right|$$

Now, $\frac{1}{2} \left| -5(-5-(-6) + -4(-6-(7) + (-1)(7-(-5) \right|$

$$= \frac{1}{2} \left| -5(1) + (-4)(-13) + (-1)(12) \right|$$

$$= \frac{1}{2} \left| -5 + 52 - 12 \right| = \frac{1}{2}|35| = 17.5 \text{ units square}$$

Now for ΔACD

$$\frac{1}{2} \left| -5(-6-(5) + (-1)(5-(7) + 4(7-(-6) \right|$$

$$\frac{1}{2} \left| -5(-11) + (-1)(-2) + 4(13) \right|$$

$$\frac{1}{2} \left| 55 + 2 + 52 \right| = \frac{1}{2} |109| = 54.5 \text{ units square}$$

So, Total area is 72 sq units.

30. HCF of 861 and 1353 :-
1353 = 861 × 1 + 492
861 = 492 × 1 + 369
492 = 369 × 1 + 123

$369 = 123 \times 3 + 0$
So, HCF of 861 and 1353 is 123.

31. A point of x-axis means y-co-ordinate is 0
So it is in form of $(x, 0)$
Given
Distance of $(x,0)$ from $(5,-2)$ and $(-3,2)$ is equal
By distance formula
$\Rightarrow \sqrt{(5-x)^2 + (-2-0)^2} = \sqrt{(-3-x)^2 + (2-0)^2}$
$\Rightarrow \sqrt{(5-x)^2 + 4} = \sqrt{(-3-x)^2 + 4}$
$\Rightarrow (5-x)^2 + 4 \; (-3-x)^2 + 4$
$\Rightarrow (5-x)^2 = (-3-x)^2$
$\Rightarrow \cancel{5-x = -3-x} \Rightarrow 25 + x^2 - 10x = 9 + x^2 + 6x$
$= 16 = 16x \Rightarrow x = 1$
So, value of x is 1

Section D
32. $\tan A + \sin A = m$
$\tan A - \sin A = n$
$m^2 = (\tan A + \sin A)^2 = \tan^2 A + \sin^2 A + 2 \sin A \tan A$
$n^2 = (\tan A - \sin A)^2 = \tan^2 A + \sin A - 2 \sin A \tan A$
$m^2 - n^2 = (\tan A + \sin A)^2 - (\tan A - \sin A)^2$
$= 4 \sin A \tan A$ --- (i)
Now, $mn = (\tan A + \sin A)(\tan A - \sin A)$
$= \tan^2 A - \sin^2 A$
$= \sin^2 A / \cos^2 A - \sin^2 A = \dfrac{\sin^2 A - \sin^2 \cos^2 A}{\cos^2 A}$

$\dfrac{\sin^2 A(1 - \cos^2 A)}{\cos^2 A} \Rightarrow \dfrac{\sin^2 A(\sin^2 A)}{\cos^2 A}$ $\{1 - \cos^2 \theta = \sin^2 \theta\}$

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DOMS Page No.
Date / /

$\frac{\sin^4 A}{\cos^2 A} = mn$ and root $mn = \frac{\sin^2 A}{\cos A}$ --- (ii)

from (1)

$\sin A \cdot \tan A = \sin A \times \sin A / \cos A = \sin^2 A / \cos A$ -- (i

So, LHS

$= 4 \sin A \cdot \tan A \Rightarrow 4\sqrt{mn}$ from (iii)

Hence Proved

23. Given

33. Given

$x-y+1=0$ --- (i)

$3x+2y-12=0$ --- (ii)

Now,

$x-y=-1$ and $3x+2y=12$

for the (i) equation.

$x=-1+y \Rightarrow x=y-1$

x	-1	0	1	2	3
y	0	1	2	3	4

for the (ii) equation

$3x+2y=12 \Rightarrow 3x=12-2y \Rightarrow x=4-2/3y$

x	4	10/3	8/3	2
y	0	1	2	3

x, y intersects at 2,3

Points on x-axis are: (-1,0) and (4,0)

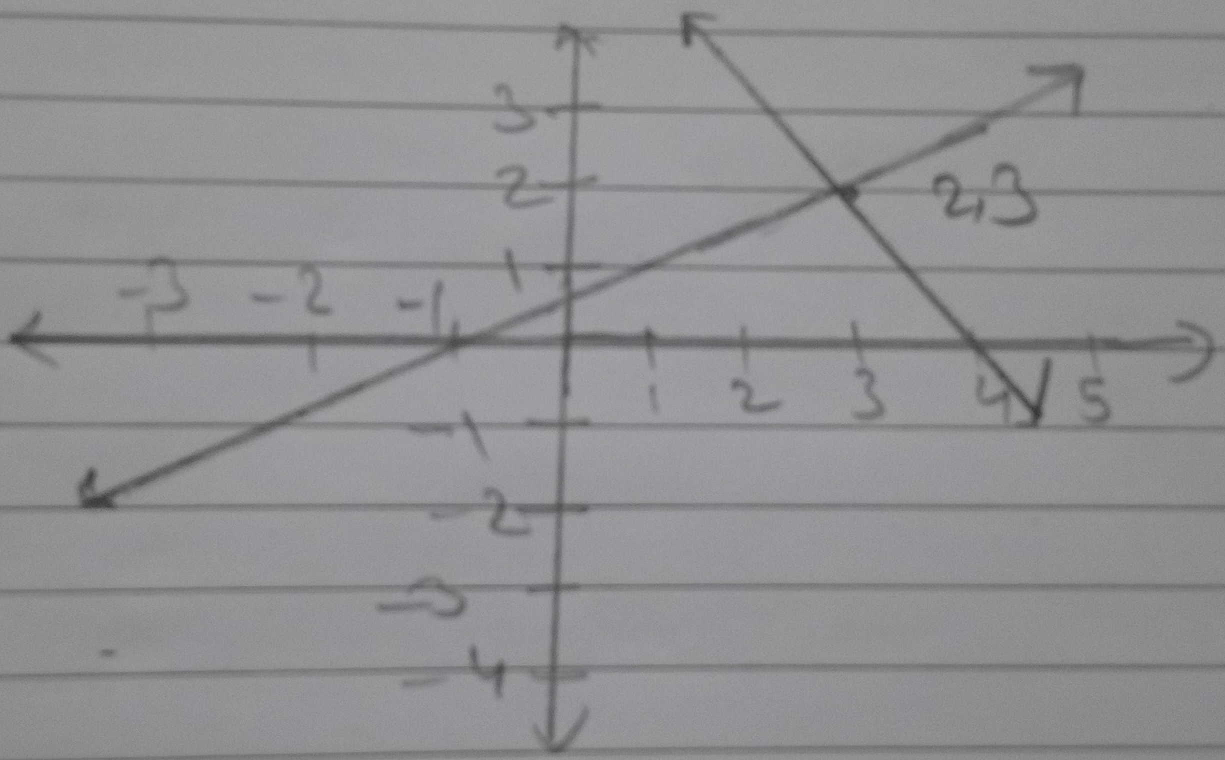
So the area of triangle formed by it is :-

$\frac{1}{2}\left|x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\right|$

$\frac{1}{2}\left|-1(3-6)+2(0-0)+4(0-3)\right|$

$\frac{1}{2}\left|(-3)+0+(-12)\right| = \frac{1}{2}\left|-15\right| = \frac{1}{2}(15)$

Area of triangle formed by intersection and x-axis is 7.5 sq units.


3
2
1
-1
-2
-3
-4
-3
-2
-1
1
2
3
4
5
2,3

34. Given:

polynomial = $(2x^3 - 4x - x^2 + 2)$

Two zeros are $\sqrt{2}$ and $-\sqrt{2}$

So,

we can write

$x - \sqrt{2} = 0$ and $x + \sqrt{2} = 0$ as they are zero/roots

Now, $(x-\sqrt{2})(x+\sqrt{2}) = x^2 - 2$

Now,

$$x^2 - 2 \overline{)\, 2x^3 - 4x - x^2 + 2 \,}\, (2x - 1$$

$$- \underline{2x^3 - 4x}$$

$$-x^2 + 2$$

$$- \underline{-x^2 + 2}$$

$$0$$

So, ~~is also a zero~~

$(2x-1)(x-\sqrt{2})(x+\sqrt{2}) = 2x^3 - 4x - x^2 + 2$

Now, $2x - 1 = 0 \Rightarrow 1 = 2x \Rightarrow 1/2 = x$

Therefore it's all zeros are

$1/2, \sqrt{2}, -\sqrt{2}$

Since it has degree of 3 it have 3 zeros.

35. Given:

Point $A = (4, -3)$

Point $B = (8, 5)$

Ratio is which they divide 3:1

We need to find point C (Point which divides the line segment in 3:1)

So,

By Section formula

$$\left[\left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}\right), \left(\frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}\right)\right] = C(x, y)$$

$m_1 = 3$, $m_2 = 1$, $x_1 = 4$, $x_2 = 8$, $y_1 = -3$, $y_2 = 5$

Now,

$$\left[\left(\frac{3(8) + (1)(4)}{3+1}\right), \left(\frac{3(5) + 1(-3)}{3+1}\right)\right] = \text{Point } C\ (x, y)$$

$$= \left[\left(\frac{24+4}{4}\right), \left(\frac{15-3}{4}\right)\right] = C(x,y)$$

$$= [7, 3] = C(x,y)$$

So the co-oridnates for C are (7,3)

Section E

36.

(i) 2) (0, 12)

(ii) 1) (3,5)

(iii) 3) 5m

37.

(i) 3) Parabola

(ii) 3) 0

(iii) 3) -3,2

38.

(i) 3) $24 cm^2$

(ii) 1) RHS

(iii) 4) Converse of Pythagoras Theorem